



MAGE: Alleviating Uncertainty in Real-Time Decision-Making as a Function of Problem Complexity

Natalia M. Alexandrov¹

NASA Langley Research Center, Hampton, Virginia 23681-2199, USA

In this paper, we discuss a critical aspect of uncertainty in the operation of complex systems, such as the future air traffic: the ability of agents in the system to arrive at satisfactory decisions and the attendant actions as a function of problem complexity. Intuitively, when the problem complexity is manageable, given an appropriate decision problem formulation and solution tools, an agent (computational or human) has no trouble arriving at a solution that yields good outcomes for the agent and the system. Growing problem complexity results in progressively larger computational problems that may yield suboptimal solutions or even be intractable within required time limits or at all. We propose a measurable representation of complexity in terms of problem tractability and quality of solutions. We also propose a computational scheme, MAGE (Monitor, Anticipate, Guide, Evolve), for detecting approaching transitions from efficient decision-making states to inefficient to unsafe ones, so that operations based on decision-making can be reconfigured to forestall unfavorable transitions, returning to efficient modes when complexity diminishes. Maintaining tractable complexity reduces the uncertainty in the outcomes of decision-making. We describe the general scheme, an outline of MAGE applied to managing airspace complexity, and initial examples of investigating the tractability of problem-solving schemes.

I. Introduction

In this paper, we address a major source of uncertainty in the design and operation of complex systems – the ability of participating agents to solve problems in dynamic, time-critical problem-solving environments.

To illustrate the concept, let us consider an example of aircraft motion in a future airspace. In perfect weather, with no other aircraft or restricted areas, the optimal trajectory in cruise would lie along the geodesic connecting the vehicle's origin to destination. With other aircraft, static and moving obstacles, restricted areas, or inclement weather present, the vehicle would, instead, compute deconflicted paths toward reaching the destination, by solving decision-making problems at specified time intervals, while attempting to maintain as efficient performance as possible.

The decision-making problems include constraints on the physical attributes of the aircraft (e.g., speed, maneuverability, fuel) and interaction with other airspace participants and events (weather, the number of aircraft close enough for potential conflicts, their speeds, intentions, and other variables), all with a degree of uncertainty. The problem includes several objectives, such as minimal distance traveled, delay, and terms intended to moderate selfish behavior at the expense of the overall system performance.

In sparse airspace, computing a conflict-free path in time is easy. In dense and heterogeneous traffic, the problem is larger, has more uncertainty about the system properties, and may yield an inefficient or even an unsafe path. Sometimes, to avoid a conflict, the aircraft moves away from its destination. As density and heterogeneity grow, the aircraft finds itself in Brownian motion: instead of moving toward destination, it is only avoiding collisions.

An early analysis by Alexander [1] showed via a particle model, that a randomly distributed system that starts with hours between two-body interactions and 1000 hours between three-body interactions transitions to 20 minutes between interactions when the density increases ten-fold and transitions to nearly continual maneuvering with another ten-fold increase in density. Although the system was idealized in that example (e.g., there was no accounting for heterogeneity of participants), the trend and the numbers are highly instructive. In short, the airspace becomes complex quickly with increasing density, which presents a difficult computational problem for arriving at safe trajectories.

¹ Principal Investigator; Autonomous Integrated Systems Research Branch; AIAA Associate Fellow.

More recent probabilistic estimates of conflicts and collisions in various environments can be found in work by, e.g., Arnaldo et al. [2] and Paielli and Erzberger [3, 4], with an extensive bibliography on rare collisions provided by Holzmann [5]. In particular, the authors of [3] and [4] use probabilistic estimates of conflict for aircraft in free and non-level flight to propose a theoretical foundation and a computational tool for assisting air traffic controllers in maintaining safe airspace, developed for the current control environment. An estimate of conflict probability is one measure of airspace complexity that must be bounded.

Complexity is bounded now, in the contemporary and very safe airspace controlled by humans. However, the complexity bounds are static and are related to the cognitive capacity of a human controller, i.e., the limits of how many aircraft a human can manage simultaneously. Sridhar et al. [6] and Kopardekar et al. [7] discuss measures of complexity in the context of the present airspace.

Arguably, the projected future airspace will possess an unprecedented degree of density and heterogeneity that will necessitate the transfer of control to human-machine teams and entirely autonomous algorithms. Regardless of whether the decision maker is a human or an algorithm, the general principle remains: the problems must not be more complex than the capabilities of the problem solver. This, again, translates to bounding complexity. The question is how to bound complexity in a dynamic and scalable way amenable to computational decision-making.

In this paper, we introduce a branch of the *ab initio* approach to system design and operations outlined in [8]. Because future airspace operations and the flight of individual aircraft will rely heavily on algorithmic decision-making, we pursue the following objectives:

- Overall: Prevent the system from descending into inefficient, then unsafe decision-making states where the uncertainty about good outcomes of decisions is high, i.e., keep the system sufficiently simple.
- Represent complexity explicitly in terms of computable decision-making characteristics of the algorithms that control operations, e.g., the motion of aircraft from origin to destination. While human decision-making contains many implicit and not well-understood variables, machine decisions must be formalized and understood explicitly.
- Detection of an approach to an onset of phase transition from efficient and safe to inefficient or uncontrollable operations must take place in time to forestall an onset of inefficient or unsafe system operation via reconfiguration of operations.
- The proposed scheme should make no assumptions on the overall control and decision strategy (centralized, distributed, hybrid) *a priori*, including who or what controls the decisions and reconfiguration in the system. Instead, appropriate control strategies should “fall out” of the airspace parameters and state of the art in relevant technologies. For instance, the uncertainty related to the quality of communications and the presence of equipment may allow various degrees of control autonomy to different classes of aircraft. A range of reconfiguration strategies may be encoded, or individual aircraft may make reconfiguration decisions autonomously or in groups.
- In a trade-off between pursuing individual objectives vs. maintaining the overall system safety and efficiency, we wish to impose as few constraints as possible on individual system participants.

To achieve these objectives, we propose the following:

- A computable measure of complexity consists of two output variables:
 - Tractability, with a look ahead, of the solution problem on a time budget, as a function of the external and internal parameters. In the case of air traffic, the external parameters include the density and heterogeneity of the relevant airspace volume, while internal parameters include the physical properties of the aircraft and computational properties of decision-making algorithms, among others.
 - The quality of solutions of the decision-making problem can be measured in terms of constraint satisfaction, optimality, and robustness. In the case of air traffic, one measure of robustness is in terms of size of the feasible space around a solution, i.e., robust solutions live in the regions of space that allow for many alternative solutions. One notion of such robustness is flexibility preservation (e.g., [9]), where a solution is chosen from a discrete set of solutions to allow for a larger set of subsequent solutions (e.g., [9]).
- Because, in general, state-of-the-art decision-making algorithms, such as nonlinear optimization, lack *a priori* results for global convergence rates and quality of solution on a time or iteration budget, tractability and quality of solution can be approximated via data fitting models, such as machine learning. In the case of the airspace problem, training would be done on large numbers of solutions for

path computation scenarios. These models can then be used to anticipate approaching phase transitions in the normal ebb and flow of traffic, as well as for detecting anomalous conditions. Once an approach to transition is detected, control would be reconfigured to avoid transition, returning to nominal configuration once complexity decreases. Reconfiguration would be achieved by changing the formulation of the decision-making problem and would depend on the type of the decision-making algorithm.

In the remainder of the paper, we describe the concept of MAGE (Monitor, Anticipate, Guide, Evolve) that we believe will accomplish these objectives. MAGE monitors environmental complexity at multiple scales, predicts the approach of deterioration in operations, and guides dynamic problem-solving reconfiguration to dissipate operational bottlenecks, thus maintaining an environment conducive to efficient and safe operations.

II. Conceptual Approach and Algorithm

The specifics of complexity management for problem solving depends on the problem context. However, because the overarching approach is applicable to any setting, we first describe the method via a general pseudocode. In the subsequent section, we formulate the pseudocode and its elements in terms of motion of an agent through airspace.

Algorithm 1: Conceptual Complexity Management for a Problem-Solving Agent

Given:

An agent and a goal

Initialize:

Set time $t=0$

Set initial sampling time interval Δt

Set initial look-ahead time T

Set initial look-ahead sampling time interval $\Delta \tau$

Select initial problem-solving algorithm P

Select environmental complexity parameters C

Select initial environmental complexity model M

Select transition criteria

Select stopping criteria

Do until (stopping criteria are satisfied)

1. *Acquire and assess complexity parameters C at time t*

2. *Set $\tau = t$*

Do while ($\tau \leq t + T$)

Estimate look-ahead complexity parameters at time τ

$\tau = \tau + \Delta \tau$

End do

Input complexity parameter array to complexity model M ; assess approach to phase transition

3. *If (approach to transition detected) then*

Update Δt , $\Delta \tau$, T

Update M

Reconfigure operations toward goal

Else

Assess system's performance slack

If (slack)

Reconfigure operations toward goal

Else

Continue present operations toward goal

End if

End if

$t = t + \Delta t$

End do

The pseudocode describes common-sense monitoring and management of complexity. At a given sample interval, an agent finds out what decision-making problems it will have to solve during several projected steps. If solving the problems on available time and computational budget is unlikely, the agent reduces the solution difficulty by changing its relationship to the environment.

The assessment of the problem solvability may be done by pre-solving, given sufficient resources. But we make no assumptions about the ability to pre-solve and, instead, rely on a complexity model.

The complexity model M , based on data fitting, connects the environmental parameters, such as the number of interacting agents and their properties in the system with the time required to solve the decision-making problems to proceed with operations and the problem-solving quality. Problem solving via explicit optimization algorithms lends itself to such modeling particularly well.

Reconfiguration is accomplished by modifying objectives, constraints, and variables in the decision-making problem. For instance, optimality requirements may be relaxed in favor of assigning fixed values to some of the variables.

Note that the approach to phase transition in terms of the problem solution parameters and the corresponding tractability and quality of solutions may be gradual or abrupt. While we would like there to be little time difference between solving a problem with five variables and a problem with six variables, such a conclusion may not be available *a priori*. However, the hope is that a sound design of experiments and massive data collection would yield a sound functional dependence of complexity metrics on complexity parameters at sufficient resolution to forestall the loss of efficiency or even solvability.

III. Application to Air Traffic

Let us consider the elements of the general approach applied to conjectured future air traffic in an arbitrary volume of space. We focus on decision-making at a single scale, that of an individual aircraft. Ultimately, the number and type of observations and decision-making scales, leading to a control architecture (e.g., distributed, centralized, hybrid), would be derived based on controllability requirements, given airspace parameters and the state of relevant technology at the time of operations. In this work-in-progress paper, however, we restrict decision-making and complexity assessment to the perspective of a single agent we will call “POV” (point of view) agent.

The MAGE concept consists of two stages: model building (or “learning”) and operation. We conjecture that formal optimization algorithms are likely to be used for path computation in future airspace. Computational complexity of general nonlinear optimization is unpredictable for a specific problem. Therefore, the learning stage of MAGE aims to recognize potential reduction in tractability based on large amounts of synthetic data. Eventually, we hope that online learning can be incorporated. In this application, we are not relying on data fitting for immediate safety-critical decisions, for which data-fitting techniques, such as machine learning, are not ready and may not be ready, in principle. Instead, we are using data fitting to recognize patterns of solvability, and MAGE would be used in an advisory capacity to the ultimate decision-maker, algorithmic or human.

A. Complexity Model Construction and Use

The current version of the complexity model under development takes the following inputs: the number of aircraft in a given volume around an agent at time t ; the velocities of the POV agent and other aircraft; the type of decision-making problem formulation for path computation; and the solution scheme used to compute deconflicted trajectories. The latter may consist of a single algorithm or a sequence of solution strategies, including heuristics. The model assumes that the velocities will persist for several given time steps within the look-ahead time T .

Given input, the model estimates the number of variables and constraints for each time step within the look-ahead temporal interval. Then, it issues a prediction of time and quality of solutions at each time step.

The prediction is based on a data-fitting model, with data on solution time and quality collected for multiple scenarios. At the time of writing, simple polynomial models and machine learning models are in construction.

Once the model is constructed, it is used at each step of path computation to inform the potential reconfiguration of airspace realized via either external direction or an autonomous algorithmic decision.

The following are examples of other considerations that would go into building a model for practical use:

- Regions of airspace with highly dissimilar nominal loads may require separate models. It is not clear now whether a single universal model should be constructed.
- There is much unknown about the degree of uncertainty about the decision-making software on board of highly heterogeneous future aircraft, what “rules of the road” the majority of aircraft would share, and what fraction of aircraft may be uncooperative, because of lack or failure of equipment or, possibly, intentional behavior. Where information is not reliably available, worst-case estimates can be

made. However, the overuse of worst-case estimates may lead to unduly conservative decisions about reconfiguration and unnecessary loss of efficiency. Incorporation of uncertainty about individual decision-making methods is an active area of investigation for MAGE.

- Once developed, a complexity model is assumed to remain unchanged, for now. Real-time learning scenarios for model updating are in development as real-time updating may be necessary in case of rapid and unexpected changes in the environment.

Many other sources of uncertainty, such as the quality of communications and information about intent enter assessing both tractability and quality of solution. However, formalized decision-making, especially via explicit optimization algorithms, lends itself well to estimating solution sensitivity to input uncertainties. Moreover, high uncertainty informs the need for larger safety buffers and other safety measures.

B. Reconfiguration

From the implementation perspective, by “reconfiguration” we mean a change in the decision problem formulation for the purpose of augmenting the goals of a vehicle’s mission and overall airspace operation. For instance, while minimizing arrival delay for an individual vehicle may be an objective with a high weight during normal, efficient operations, the delay objective may lose importance in high-risk, dense situation, necessitating a change in the path computation of that vehicle. In another example, if many vehicles attempt to converge on a large-scale event at, e.g., a sports arena, some of the free variables in the decision-making problems may be eliminated and set as fixed parameters, to enforce an organized and efficient flow of traffic, thus simplifying the problem.

Problem reformulation mechanisms also include explicit addition or subtraction of objective functions and switching the roles of objectives and constraints for some of the problem’s functions. For example, hard business cost constraints present during normal operations may transition to softer objectives when the system reconfigures to a safety mode.

Reconfiguration may be initiated in several ways, resulting in different outcomes. Problem solving tractability and quality of solutions can be observed from the perspective on an individual aircraft or a set of aircraft or a large volume of airspace. Each of the levels in multiscale observation can be endowed with decision-making authority. In fact, decision-making authority may change based on need. We conjecture that dynamic multilevel authority would lead to the safest and most efficient system. The following are examples of reconfiguration mechanisms and outcomes:

1. In moderately dense and heterogeneous airspace with cooperative participants, individual aircraft can modify their decision-making problem formulation autonomously, i.e., based on local information, without external control. The outcomes of such reconfiguration could lead to small, local shifts in overall traffic or they could propagate in an unpredicted manner and lead to large-scale changes in traffic patterns. Such “emergent” events based on local changes are under investigation in the MAGE project.
2. Growing and propagating density may mean that high-fidelity information is no longer available to individual participants in time. Even if information is available, the computational load for reconciling this information may be excessive for the resources of an individual aircraft. In that case, a medium-scale level or a large-scale level observer may direct reconfiguration of problem solving or, in extreme cases where safety is impaired, can impose trajectories on aircraft in a given volume. In such a directed reconfiguration, the outcomes would be more certain, although not necessarily more efficient.
3. Finally, a hybrid approach would allow individual aircraft control its reconfiguration while relinquishing control to an agent that has better, more global information, when justified.

C. Decision-making Strategies

How the decision-making formulation is solved is one of the independent variables that affects the outcome of actionable complexity prediction. Algorithmic performance of nonlinear optimization algorithms and other methods for finding solutions changes rapidly with developments in hardware and software. As new capabilities arise, complexity models must be re-calibrated.

IV. Initial Numerical Assessment of Predicting Tractability

Comprehensive numerical experimentation with the MAGE concept is in development and relegated to another publication. Here, we report on a first step in assessing predictability of decision-making tractability as a function of environmental complexity.

The initial set of tests concerns the movement of an object from origin to destination in 2D. We assume that the path is recomputed at specified time intervals, based on the information about the environment available by the start

of each time interval. We accumulate performance characteristics of the solution at one time step for a “training set” of cases and verify whether similar times result in the test cases outside the training set, without building a formal model yet.

In practical situations, the single-step computation will contain near-term temporal and spatial constraints (e.g., conflict avoidance), as well as far-term effects of the accumulated near-term decisions. For instance, avoiding a few potential conflicts while minimizing the deviation from the optimal unconstrained trajectory may not result in a large increase of distance traveled. However, in the presence of many potential conflicts, avoiding each one while staying as close as possible to the optimal unconstrained path may incur more extra distance than avoiding the area altogether, which would increase the deviation from the ideal path at each step but result in a smaller overall cost in, e.g., fuel expenditure.

In the initial tests, we do not include such details. Rather, the aim is to detect trends in computational tractability of simple decision problem formulations, using examples of two formulations and two decision-making schemes.

We consider decision-making performed by a single software POV agent X at a single time step. The number of other agents in the area under consideration represents environmental complexity. The obstacles are placed randomly in relation to agent X.

In the present tests, we do not yet consider heterogeneous agents. In subsequent tests, varying velocities will stand for heterogeneity.

D. Shortest Path Finding via Visibility Graph

In this test, shortest path finding in a plane with circular obstacles of the same size represents a single step in 2D movement from origin to destination.

At the beginning of a computational step, agent X receives information about the current position, speed, and heading of other objects in a given area. Projecting the position of the objects at a given time interval in the future yields a “snapshot” of several circular obstacles (denoting the spatial buffer around the objects) among which agent X must compute a path without conflicts. Fig. 1 shows a typical setting for computing a path from point A to point B, with several obstacles.

We use the visibility graph (e.g., [10]) approach to compute the path (see, e.g., [11] for an overview of seminal results). Our implementation follows CodinGame programming examples².

Although the obstacles are circular, we first approximate them by regular polygons in order to generate a finite number of vertices for building the visibility graph. In a visibility graph, the nodes represent positions and the edges represent visible connections between them.

We then use the A* algorithm [12] to compute the path on the bases of the visibility graph. The implementation is basic, with no attempts to improve the heuristic. Table 1 summarizes the results.

For each number of obstacles, one hundred scenarios of random distribution in a square containing the straight line (optimal unconstrained path) between point A and point B. Of course, many of the obstacles are not in the direct interaction range of the path between A and B. However, our purpose in this initial experiment is to detect trends in the basic behavior of the search algorithms, and so no code optimization was done. The plans for data collection from more realistic scenarios include the use of optimized, commercial software.

The same procedure is applied to the test scenarios that would be considered as off-training data for model testing.

Our initial, naïve implementation of the algorithm resulted in a rapid decrease of the number of solutions found as a function of the number of obstacles. Nonetheless, even a naïve implementation proved to be a good predictor of the algorithm’s performance on off-training test problems, as shown in the last column of Fig. 2 – the percent difference between the average time to solution in the learning set and the average time to solution in the test set.

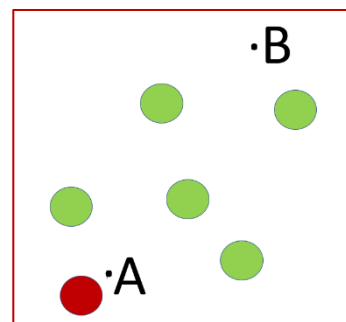


Figure 1. A typical snapshot of obstacles (green) near a POV agent (red).

² <https://www.codingame.com/start>

# obstacles	# scenarios	Avg. # solutions	Avg. time to solution (sec)	# test scenarios	Avg. # test solutions	Avg. % time prediction Δ
4	100	97	0.86	25	24	0.02
5	100	89	1.31	25	19	0.07
10	100	34	324.33	25	7	2.03
20	100	2	4800.00	25	Not found	N/A

Figure 2: Summary of performance for a visibility graph approach.

E. Minimization of Deviation from Optimal Path

In this test, the geometric setup is similar to that of the previous test, with randomly generated obstacles. The solution strategy for path computation is an idealized, explicit optimization formulation. The objective function is the difference between the position of agent X at the end of the step computation and the position on an ideal, unconstrained path (in this case, a straight line). The constraints are on exceeding the separation limits. The speed is assumed to be constant.

The formulation was solved by using the Kreisselmeier-Steinhauser method, KSOPT [13], which is in the class of sequential unconstrained minimization [14] techniques. KSOPT converts constrained problems to a sequence of unconstrained ones.

# obstacles	# scenarios	Avg. # solutions	Avg. time to solution (sec)	# test scenarios	Avg. # test solutions	Avg. % time prediction Δ
4	100	94	0.38	25	25	0.01
5	100	100	0.37	25	23	0.01
10	100	83	11.60	25	24	0.03
20	100	86	64.52	25	19	0.05

Figure 3. Performance summary for minimizing deviation from unconstrained optimal path.

Although the problem formulation was extremely idealized, the % of execution time error in test problems shows a reasonable prediction of performance obtained for the problems of the same size during the “learning” stage.

F. Discussion of Preliminary Results

To reiterate, the initial tests were not meant to compare solution methods or select a specific efficient problem-solving scheme. Rather, they served to check whether the hypothesis that tractability could be predicted as a function of problem size proved promising, at least for simple problems.

The work in progress now includes the increase in fidelity of the test problems, to assess performance prediction in more realistic scenarios. Please note that the demonstrated prediction would be less significant if we viewed performance across a range of general nonlinear programming problems. However, we expect that a single comprehensive and basic decision-making problem formulation (or, at most, a few) would be selected in practice. Thus, we are cautiously optimistic that on-a-budget performance attributes will show clear functional connection to the parameters of problem complexity. Reconfiguration will affect problem formulation, but the classes of reconfigured formulations would be limited as well.

V. Concluding Remarks

In summary, we want the system to be controllable. Control always works by bounding complexity. Simply put, the problems that arise in the system must not be more complex than the solvers can solve. In our current and very safe airspace control system, bounds are based on experience with human controller workload capacity. In the future system, the idea is the same. However, density, heterogeneity, and autonomy of the future system will require different controller teams, from human-machine teams all the way to complete machine control autonomy.

There is a close relationship between trust and control of the system. The less trust, the more control is needed. We claim that bounding system complexity is a necessary condition for making any statements about safety and certification of a trustworthy system. This requires formalizing the system at a fundamental level, so we can reason about it and derive control requirements based on that reasoning, rather than on point designs.

Our hypothesis is that to formalize a dense and heterogeneous system in a useful way amenable to reasoning about it, complexity *per se* is too fuzzy a measure. However, when we talk of complexity, we really mean solvability. This leads to a useful formalization, because every action at every scale in the system is an outcome of

decision-making, either immediate or accumulated, explicit or implicit. When we consider solvability directly, we can connect system formalization with reasoning about actions in the system and system design. Specifically, we can detect an approach to conditions that prevent reaching satisfactory solutions and modify operations before this happens.

Initial testing of performance predictability for highly simplified problems and two solution strategies indicate that further development of the planned complexity models is justified. The work on more realistic multi-agent path computing in airspace is in progress, as well as on acquisition of algorithms for solving the test scenarios. More realistic scenarios will serve as a basis for data collection to construct complexity models. Following the assessment of complexity models, we plan to evaluate various multi-agent control strategies with respect to performance and the attendant system uncertainties.

Acknowledgments

The author would like to thank the NASA Langley Research Center CIF/IRAD Program for supporting this work.

References

- [1] Alexander, B.; *Aircraft Density and Midair Collisions*, Proceedings of the IEEE, 58(3):377-381, 1970
- [2] Arnaldo, R., Sáez, F., Garcia, E. and Portillo, Y.; *Probability of Potential Collision for Aircraft Encounters in High Density Airspaces*, Intech, <http://dx.doi.org/10.5772/48685>, 2012
- [3] Paielli, R. and Erzberger, H.; *Conflict Probability Estimation for Free Flight*, J. Guidance, Control, and Dynamics, 20(3), May-June 1997
- [4] Paielli, R. and Erzberger, H.; *Conflict Probability Estimation Generalized to Non-Level Flight*, Air Traffic Control Quarterly, 7(3), 1999
- [5] Holzmann, T.; *Probabilistic Estimation of Rare Random Collisions in Space*, Thesis, AFIT/GOT/ENS/09-07, Department of the Air Force Air University, Air Force Institute of Technology, September 2007
- [6] Sridhar, B., Sheth, K., and Grabbe, S.; *Airspace Complexity and its Application in Air Traffic Management*, 2nd USA/Europe Air Traffic Management R&D Seminar, Orlando FL, 1998
- [7] Kopardekar, P., Magyartis, S.; *Dynamic Density: Measuring and Predicting Sector Complexity*, 21st Digital Avionics Systems Conference, Irvine CA, 2002
- [8] Alexandrov, N.; *Control of Future Air Traffic Systems via Complexity Bound Management*, 2013 Aviation Technology, Integration, and Operations Conference, August 12-14, Los Angeles, CA, AIAA 2013-4419, 2013
- [9] Idris, H., Vivona, R., Garcia-Chico, J.-L., and Wing, D.; *Trajectory Planning by Preserving Flexibility: Preliminary Metrics and Analysis*, DOI: 10.2514/6.2008-7406, August 2008
- [10] de Berg, M.; van Kreveld, M.; Overmars, M.; Schwarzkopf, O.; (2000); *Chapter 15: Visibility Graphs*, Computational Geometry (2nd ed.), Springer-Verlag, pp. 307–317, 2000
- [11] Lozano-Pérez, T.; Wesley, M.; *An algorithm for planning collision-free paths among polyhedral obstacles*, Communications of the ACM, 22 (10): 560–570, doi:10.1145/359156.359164, S2CID 17397594, 1979
- [12] Hart, P. E.; Nilsson, N. J.; Raphael, B.; *A Formal Basis for the Heuristic Determination of Minimum Cost Paths*. IEEE Transactions on Systems Science and Cybernetics. 4 (2): 100–107. doi:10.1109/TSSC.1968.300136, 1968
- [13] Wrenn G.; *An indirect method for numerical optimization using the Kreisselmeier–Steinhauser function*. Contractor Report NASA CR-4220, NASA Langley Research Center, Hampton, VA. 1989
- [14] Fiacco, A.V. and McCormick, G.P.; *Nonlinear Programming: Sequential Unconstrained Minimization Techniques*. John Wiley, New York, 1968